

WordBuds

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Our Sponsor

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Cognitive Science (LECS) Lab*

*Studies how language and
literacy are learned and
taught from early childhood
through adolescence*

Our Purpose

“To bring WordBuds to the app store, as well as provide a safe, user-friendly, and customizable interface for all prospective users”



Initial Scope

- Allowing users to give informed consent
- Adding customizable notifications
- Going to the play store/app store



Github Repo Overview

Mobile App

- User authentication through firebase
- Adding words, saved to cloud firestore
- Functioning updated UI
- Initial parent demographic page
- Ability to add multiple children
- Language support for english and/or spanish
- Support for multi-word phrases
- Statistics and visualizations for new words per week, per month and total

Researcher / Admin Web

- User authentication through firebase
- Ability to add new user accounts
- Ability to view signed up users and their configurations by roles
- Ability to view child word counts and new word distribution over time.
- Ability to export all child word data as a csv



Cloud Firestore



Backlog Themes

Theme 1 - Parent Engagement:

- Adding configurable notifications that remind parents to log their kids words each day - High Priority
- Adding a homescreen widget that users can click into to open the app directly

Theme 2 - Child Data Governance & User Preferences:

- Adding additional permissions screens when creating an account about how data is handled - High Priority
- Integrating audio handling for child voice clips saved on device when permissions do not allow sharing and saved in database when allowed

Theme 2 - Deployment:

- Upload the updated UI to the android app store
- Get approval to upload IOS app on the app store



Architectural Decisions



- Working on a pre-existing project, many architectural decisions have already been decided
- Firebase DB

Languages



● Dart 81.7%	● TypeScript 4.7%
● JavaScript 3.9%	● Python 2.2%
● C++ 2.2%	● CMake 1.7%
● Other 3.6%	

Definition of Done

- Easier to use - mods to UI, home screen widgets, notification settings, family sharing
- Data visualization
- App store deployment

Risks, Dependencies, and Unknowns

- Maintaining data integrity, user authentication
- Flutter SDX, Android Studio, Xcode on a mac device, version changes
- UI modifications, User feedback and suggestions, Deprecated Software



Demo



← → ↻ ⚠ Not secure wordbuds.info:8080 ☆ 🔍

Sign in

Email

Password

Log in

8:31 [Signal] [Wi-Fi] [Battery]



Sign in

Welcome to WordBuds, please sign in!

Don't have an account? [Register](#)

Email

Password

[Forgotten password?](#)

Sign in

[Grid] [List] [Image] [Settings] [Close]

1 2 3 4 5 6 7 8 9 0

q w e r t y u i o p

a s d f g h j k l

⬆ z x c v b n m ⬇

?123 , . ✓

Demo



← → ↻ ⚠ Not secure wordbuds.info:8080 ☆ 📄 🔒

Log out Admin panel ▾

Admin panel

Users

Role: Parent ▾ Sort by: Date created (new) ▾

Log out

- Mae Olsen
me.olsen5@gmail.com
- live Demo
chrisghom@proton.me
- ZZ
chrisghom@proton.me
- tp
tuck7712@gmail.com
- chris P2
cggghomrockingam@colimson.us.edu
- Tucker Portis
jportis0@gmail.com
- Chris Phomrockingam
chrisghom01@gmail.com

Create new User Account Export all words

User profile

Done

firstName: Mae
email: me.olsen5@gmail.com
twoFactorEnabled: false
lastName: Olsen
name: Mae Olsen
childId: [*1smXGu7Ajgyy7gm1d8Aq
status: active
role: parent
preferredLanguage: en
createdAt: 4/21/2026, 11:24:08 AM
emailVerified: true
updatedAt: 4/21/2026, 11:24:08 AM
surveyCompleted: true

11:39 📶 🔒

WordBuds Home Mary ▾

Mary's journey

0 total words recorded so far.

+ Log a new entry

Recent Words

No recent words yet. Add one to get started!

🏠 📖 + 📊 ⚙️